ONLINE RETAIL SALES ANALYSIS

From Data Cleaning to Exploratory Analysis and Interactive Power BI Dashboard

Final Data Analytics Case Study

Tools: Microsoft Excel • Power Query • Power BI • DAX

Project Overview

This project follows an end-to-end analytics workflow using an Online Retail transaction dataset. It covers data cleaning, exploratory data analysis (EDA), KPI development, and an interactive Power BI dashboard.

Problem Statement

Raw retail transaction data may contain duplicate records, cancelled invoices, missing customer identifiers, inconsistent text, and unsuitable data types. Without structured analysis and reporting, it can be difficult to identify leading products, valuable customers, important markets, and changes in revenue over time.

Dataset Information

The dataset was sourced from Kaggle's Online Retail dataset. The cleaned output was reported as 392,692 rows and nine columns: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country, and Revenue. Revenue was created as Quantity multiplied by UnitPrice.

Data Cleaning and Preparation

- Promoted the first row to headers and reviewed column data types.
- Removed duplicate records and cancelled invoices.
- Removed records with missing CustomerID values.
- Removed non-positive Quantity and UnitPrice values.

• Trimmed and cleaned text fields, then performed another duplicate check.
• Created Revenue = Quantity × UnitPrice and set it to a decimal data type.

Exploratory Data Analysis

Excel PivotTables and Pivot Charts were used to examine monthly revenue, top products by revenue and quantity, revenue by country, top customers by revenue, and unusually large transactions.

Metric	Previously Calculated Value
Total Revenue	8,887,208.89
Total Quantity Sold	5,152,002
Distinct Orders	18,532
Distinct Customers	4,338
Distinct Product Codes	3,665
Countries	37
Average Order Value	479.56

Interactive Power BI Dashboard

The dashboard includes KPI cards for total revenue, quantity, orders, customers, products, and average order value. It also includes a monthly revenue trend, Top 10 countries by revenue, Top 10 customers by revenue, Top 10 products by revenue, and Top 10 products by quantity. Slicers are provided for country, invoice date, and product description.

Key Findings

• The UK was previously calculated to contribute approximately 7,285,024.64, around 82% of total recorded revenue.
• November 2011 was previously calculated at approximately 1,156,205.61 in revenue, the highest complete month in the analysis.
• Paper Craft, Little Birdie was previously reported at approximately 168,469.60 in revenue.
• Customer 14646 was previously reported at approximately 280,206.02 in revenue.
• A transaction with 80,995 units of Paper Craft, Little Birdie was flagged for investigation as a potential anomaly or bulk purchase.
• December 2011 contains records only through December 9 and is not a complete month.

Business Recommendations

• Investigate the drivers of the UK's revenue contribution and assess opportunities in other markets.

• Review products and customer activity behind November's revenue peak. • Use product findings to inform inventory and merchandising reviews alongside margin and stock availability data.

• Examine high-value customers' purchasing patterns for retention opportunities.

• Investigate unusually large transactions before treating them as data errors.

• Compare monthly performance using complete periods or clearly label partial months.

Limitations

Revenue is not profit because costs and margins are not included. Large quantities are potential anomalies, not confirmed errors. The analysis identifies patterns but does not establish their causes. December 2011 is incomplete. Validate all KPI values against the final Power BI model before submission.

Project Links

Cleaned dataset:
https://docs.google.com/spreadsheets/d/11Ve6e2dgpOYlRJKvNHJAu6LVyuinpAJb/edit?usp=drivesdk&ouid;=10991387858 9781834118&rtpof;=true&sd;=true